THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 01

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A point lies on the x-axis. Its y-coordinate is: 1

(a) 1 **(b)** equal to its x-coordinate **(c)** negative **(d)** 0

2. The reflection of the point $(5, 7)$ in the y-axis is: 1

(a) $(-5, -7)$ **(b)** $(5, -7)$ **(c)** $(-5, 7)$ **(d)** $(7, 5)$

3. $M(-1, 4)$ is the midpoint of AB and $A(5, 6)$. The coordinates of B are: 1

(a) $(-7, 2)$ **(b)** $(4, 10)$ **(c)** $(-7, -2)$ **(d)** $(-6, -2)$

4. The value of the linear polynomial $2x + 3$ at $x = -6$ is: 1

(a) -15 **(b)** 9 **(c)** -7 **(d)** -9

5. A linear growth is represented by a straight line with: 1

(a) positive slope **(b)** undefined slope **(c)** zero intercept **(d)** negative slope

6. The number $0.\overline{27}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{41}{110}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{3}{11}$ **(d)** $\frac{11}{3}$

7. The decimal expansion of $\frac{7}{20}$ is: 1

(a) terminating **(b)** non-terminating repeating **(c)** non-terminating non-repeating **(d)** cannot be determined

8. Which of the following has a non-terminating repeating decimal expansion? 1

(a) $\frac{7}{25}$ **(b)** $\frac{3}{20}$ **(c)** $\frac{1}{7}$ **(d)** $\frac{1}{8}$

9. Expand $(5x + 2)^2$: 1

(a) $25x^2 + 10x + 4$ **(b)** $25x^2 + 20x + 4$ **(c)** $25x^2 + 20x - 4$ **(d)** $25x^2 + 4$

10. Factorise $x^2 - x - 42$: 1

(a) $(x + 6)(x + 7)$ **(b)** $(x - 7)(x - 6)$ **(c)** $(x - 7)(x + 6)$ **(d)** $(x - 6)(x + 7)$

11. ABCD is a cyclic quadrilateral with $\angle A = 65°$. Then $\angle C$ is: 1

(a) $58°$ **(b)** $115°$ **(c)** $65°$ **(d)** $295°$

12. A perpendicular drawn from the centre of a circle to a chord: 1

(a) is parallel to the chord **(b)** bisects the chord **(c)** is equal to the chord **(d)** is equal to the radius

13. Through three non-collinear points the number of circles that can be drawn is: 1

(a) two **(b)** none **(c)** infinitely many **(d)** exactly one

14. The ratio of the circumference of a circle to its diameter is: 1

(a) equal to the radius **(b)** different for different circles

(c) the same for all circles (π) **(d)** equal to 2

15. The perimeter of a circle is 88 cm. Taking $\pi = \frac{22}{7}$, its radius is: 1

(a) 28 cm **(b)** 15 cm **(c)** 7 cm **(d)** 14 cm

16. By Heron's formula, the area of a triangle with sides 3 cm, 4 cm and 5 cm is: 1

(a) 4 cm^2 **(b)** 12 cm^2 **(c)** 18 cm^2 **(d)** 6 cm^2

17. A letter is picked at random from the word 'MATHEMATICS'. The probability of picking the letter A is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{3}{11}$ **(c)** $\frac{2}{11}$ **(d)** $\frac{9}{11}$

18. If $P(E) = 0.35$, then the probability that E does not happen is: 1

(a) 0.65 **(b)** 0.55 **(c)** 1.35 **(d)** 0.35

19. If $u_1 = 1$ and $u_n = 2u_{n-1} + 3$, then u_4 is: 1

(a) 29 **(b)** 15 **(c)** 27 **(d)** 13

20. Which term of the sequence $t_n = 5n - 3$ is 607? 1

(a) $n = 122$ **(b)** $n = 121$ **(c)** $n = 124$ **(d)** $n = 123$

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. Write the coordinates of the vertices of a rectangle with one vertex at the origin, length 7 units along the x-axis and breadth 4 units along the y-axis (in Quadrant I). 2

22. An auto-rickshaw fare is ₹25 for the first 2 km and ₹15 for every further km. Find the fare for 10 km. 2

23. Find the distance between $-\frac{1}{2}$ and $\frac{5}{2}$ on the number line. 2

24. Evaluate 119^2 using $(a+b+c)^2$. 2

25. A circle has radius 13 cm. Two chords of length 10 cm each are drawn. Find their distance from the centre and state the relation between the distances. 2

26. In $\triangle ABC$ of area 48 cm^2, AD is a median. Find the area of $\triangle ABD$ and give the reason. 2

27. A box has 20 bulbs of which 3 are defective. One bulb is picked at random. Find the probability that it is 2

(i) defective

(ii) not defective.

28. How many black triangles are there at Stages 4 and 5 of the Sierpiński triangle? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Find the coordinates of the fourth vertex of a parallelogram $ABCD$ with $A(1,\,2)$, $B(4,\,3)$, $C(6,\,6)$. 3

30. The volume of a box of length 7 cm and breadth 11 cm is a linear polynomial in the height h. Find the volume for $h = 5, 9, 13$. 3

31. Verify the distributive law $\frac{1}{2} \times \left(\frac{1}{4} + \frac{1}{3}\right) = \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{3}$. 3

32. Factorise $r^2 - r - 42$ and $x^2 + 7x + 12$. 3

33. Find the area of a trapezium with parallel sides 12 cm and 20 cm and distance 7 cm between them. 3

34. One month is chosen at random. Find the probability that it has exactly 30 days. 3

35. Find the centre and radius of the circle through $(0, 0)$, $(6, 0)$ and $(0, 8)$. 3

OR

Prove that equal chords of a circle subtend equal angles at the centre. 3

36. Write the first ten terms of the Virahānka–Fibonacci sequence ($V_1 = 1, V_2 = 2$). 3

OR

Find the first four terms of $s_1 = 3,\ s_n = s_{n-1}(s_{n-1} - 1)$. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Find the distance between -4 and 3, and between $-\frac{5}{2}$ and $\frac{7}{2}$, on the number line.
(ii) Write $\left|-\frac{7}{9}\right|$, $|0|$ and $\left|\frac{5}{3}\right|$.
(iii) Between which two consecutive integers does $\frac{13}{4}$ lie? 5

OR

Show that $0.\overline{142857} = \frac{1}{7}$ and verify the cyclic property by computing 2×142857 and 3×142857. 5

38. **Case study:** A round table has centre O and radius 60 cm. Three diners sit at A, B, C on its edge with $\angle BAC = 40^\circ$. 5
(i) Find $\angle BOC$.
(ii) Find $\angle OBC$.
(iii) A fourth diner sits at D on the same arc as A; find $\angle BDC$.
(iv) If M is the midpoint of BC and $OM = 36$ cm, find BC.

OR

A cyclic quadrilateral has sides 5, 5, 12, 12 (in order). 5
(i) Show that its diagonal is a diameter and find the radius.
(ii) Find its area.
(iii) Explain why this quadrilateral is not a rectangle.

39. (i) Prove that the two paths from P to Q — one a semicircle and the other made of n semicircles on adjacent parts of PQ — have equal length. 5
(ii) Hence find the length of each path if $PQ = 28$ cm and the smaller semicircles have diameters 7, 7, 14 cm. ($\pi = \frac{22}{7}$)

OR

(i) List the integer-sided rectangles of perimeter 40 units and find their areas. 5
(ii) Which has the greatest area and which the least?
(iii) What do you notice about the shape of the rectangle of greatest area?

40. (i) Show that $25 + 26 + \cdots + 58 = 1411$ using $S_{58} - S_{24}$. **5**

(ii) Check the answer using ‘average of first and last × number of terms’.

(iii) Who first recorded this result and where?

OR

A sequence has $t_1 = -5$ and $t_{n+1} = t_n + 3$. **5**

(i) Write the first five terms.

(ii) Find the explicit rule.

(iii) Is 52 a term?

(iv) Find the sum of the first 20 terms.

END OF QUESTION PAPER **Total: 80 Marks**

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